Qualitative examples: native Prefill, decode-only visual access (posthoc, includes harms)

COCO 514668



Full [C 1.28; S 0.560]: Two cats are sitting in the back seat of a car, looking out the window.

Decode None [C 0.94; S 0.190]: Two cats are sitting in a chair.

L0 Only [C 0.94; S 0.190]: Two cats are sitting in a chair.

Early [C 0.24; S 0.160]: Two cats are sitting in a room, one is on a chair and the other is on a table.

Middle [C 2.06; S 0.500]: Two cats are sitting in the back seat of a car.

Late [C 0.40; S 0.160]: Two cats are sitting in a car, one is on the left and the other is on the right.

Early: largest SPICE degradation

COCO 381563



Full [C 1.67; S 0.167]: A plate of food with rice, vegetables, and meat.

Decode None [C 0.28; S 0.222]: A plate of food is on a table, and there is a fork and a knife on the table as well.

L0 Only [C 0.67; S 0.216]: A plate of food is on a table, with a fork and a knife placed next to it.

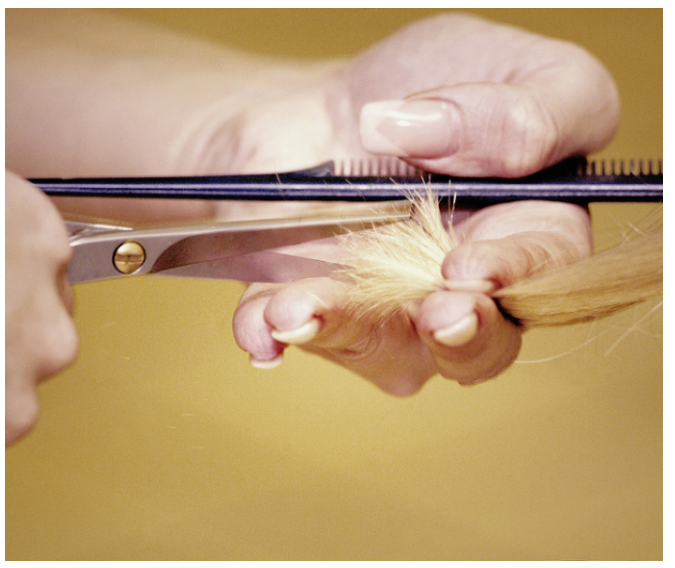
Early [C 0.90; S 0.308]: A plate of food with a variety of dishes, including a sandwich, sits on a table.

Middle [C 1.67; S 0.167]: A plate of food with rice, meat, and vegetables.

Late [C 0.62; S 0.216]: A plate of food is on a table, with a fork and a knife placed on the plate.

Early: largest SPICE improvement / least harm

COCO 401024



Full [C 2.32; S 0.296]: A person is cutting hair with a pair of scissors.

Decode None [C 0.50; S 0.000]: A person is cutting a cake with a knife.

L0 Only [C 0.50; S 0.000]: A person is cutting a cake with a knife.

Early [C 2.32; S 0.296]: A person is cutting hair with a pair of scissors.

Middle [C 2.32; S 0.296]: A person is cutting hair with a pair of scissors.

Late [C 0.50; S 0.000]: A person is cutting a cake with a knife.

Late: largest SPICE degradation

COCO 91080



Full [C 0.94; S 0.200]: A kitchen with a stove, sink, and microwave.

Decode None [C 0.93; S 0.364]: A kitchen with a stove and a sink, and a refrigerator in the background.

L0 Only [C 1.31; S 0.276]: A kitchen with a stove and a sink.

Early [C 0.79; S 0.129]: A kitchen with a white counter and a white refrigerator.

Middle [C 1.23; S 0.267]: A kitchen with a stove top oven and a microwave.

Late [C 1.02; S 0.375]: A kitchen with a stove and a sink, and a red brick wall.

Late: largest SPICE improvement / least harm

COCO 455343



Full [C 0.98; S 0.160]: A bathroom with a toilet and a glass shower door.

Decode None [C 1.30; S 0.174]: A bathroom with a toilet and a sink.

L0 Only [C 1.24; S 0.174]: A bathroom with a toilet and sink.

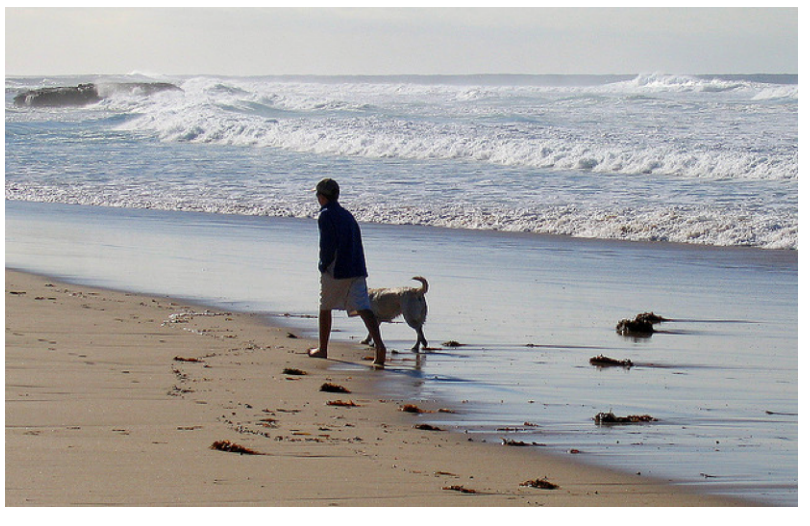
Early [C 1.30; S 0.174]: A bathroom with a toilet and a sink.

Middle [C 1.12; S 0.333]: A bathroom with a toilet and a shower stall.

Late [C 1.30; S 0.174]: A bathroom with a toilet and a sink.

Middle: largest SPICE improvement / least harm

COCO 361052



Full [C 3.61; S 0.600]: A man and his dog are walking on the beach.

Decode None [C 1.82; S 0.300]: A man is walking his dog on a leash.

L0 Only [C 1.82; S 0.300]: A man is walking his dog on a leash.

Early [C 1.82; S 0.300]: A man is walking his dog on a leash.

Middle [C 3.61; S 0.600]: A man and his dog are walking on the beach.

Late [C 2.43; S 0.500]: A man is walking on a beach with his dog.

Early minus Random10-seed02 SPICE counterexample